

Lab 18: Pertussis and the CMI-PB Project

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Investigating Pertussis Cases By Year

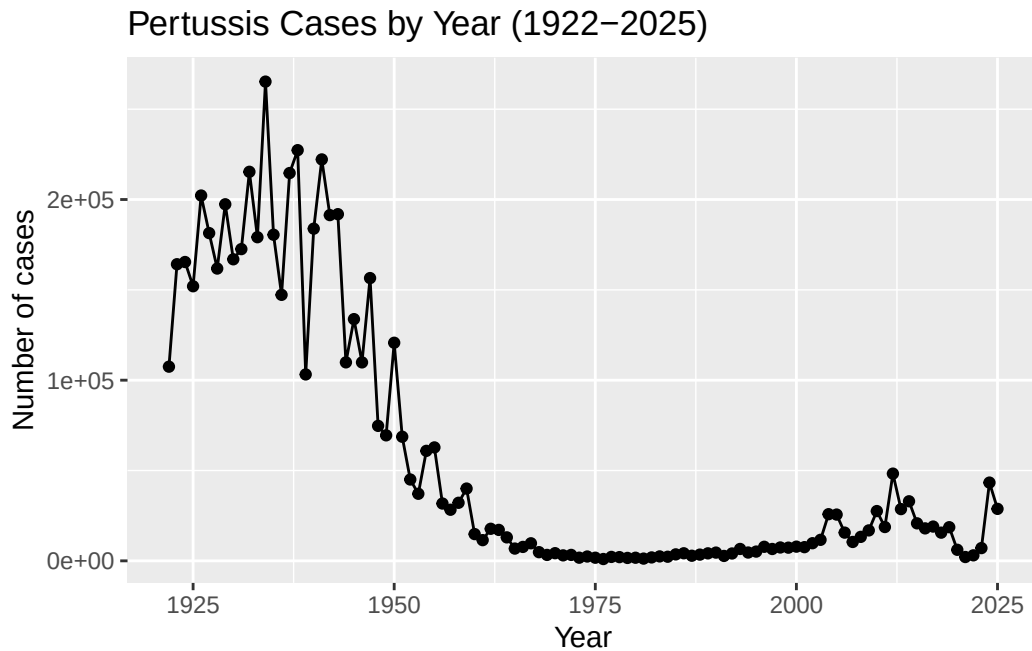
```
library(datapasta)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.2      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ggplot2)
```

```
pertussis <- read.csv("U.S. Reported Pertussis Cases_ 1922 - 2025.csv")
cdc <- data.frame(pertussis) %>%
  mutate(cases=as.numeric(gsub(",", "", .$Number.of.Reported.Pertussis.Cases)))
```

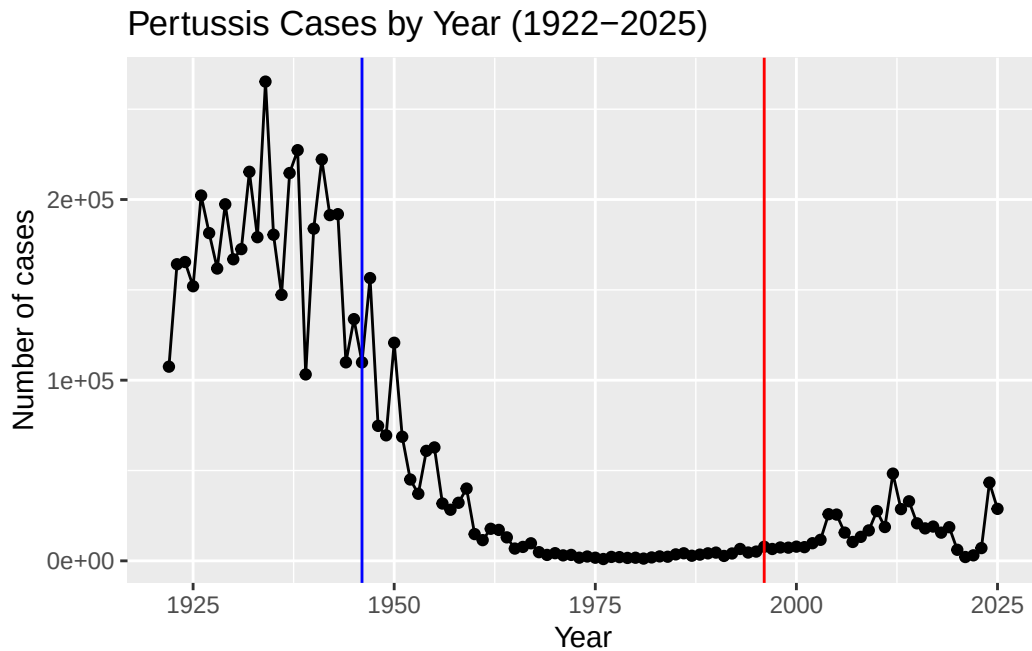
```
ggplot(cdc) +
  aes(x=Year, y=cases) +
  geom_point() +
  geom_line(aes(group = 1)) +
  labs(x="Year", y="Number of cases", title = "Pertussis Cases by Year (1922-2025)")
```



A tale of two vaccines (wP & aP)

Q2. Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

```
ggplot(cdc) +
  aes(x=Year, y=cases) +
  geom_point() +
  geom_line(aes(group = 1)) +
  labs(x="Year", y="Number of cases", title = "Pertussis Cases by Year (1922-2025)") +
  geom_vline(xintercept = 1946, color = "blue") + geom_vline(xintercept = 1996, color = "red")
```



After the introduction of the wP vaccine, the number of pertussis cases drops dramatically.

Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

After the introduction of the aP vaccine, pertussis cases seem to rise again. A possible explanation may be the bacteria developing immunity.

Exploring CMI-PB data

```
library(jsonlite)
```

```
Attaching package: 'jsonlite'
```

```
The following object is masked from 'package:purrr':
```

```
flatten
```

```
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
head(subject, 3)
```

```

  subject_id infancy_vac biological_sex ethnicity race
1          1          wP      Female Not Hispanic or Latino White
2          2          wP      Female Not Hispanic or Latino White
3          3          wP      Female      Unknown White
  year_of_birth date_of_boost      dataset
1  1986-01-01   2016-09-12 2020_dataset
2  1968-01-01   2019-01-28 2020_dataset
3  1983-01-01   2016-10-10 2020_dataset
```

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

```
table(subject$infancy_vac)
```

```

aP wP
87 85
```

Q5. How many Male and Female subjects/patients are in the dataset?

```
table(subject$biological_sex)
```

```

Female   Male
  112     60
```

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

```
table(subject$biological_sex, subject$race)
```

```

      American Indian/Alaska Native Asian Black or African American
Female                0      32                2
Male                  1      12                3

      More Than One Race Native Hawaiian or Other Pacific Islander
Female                15                1
```

Male	4	1
Unknown or Not Reported White		
Female	14	48
Male	7	32

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```
subject$age <- today() - ymd(subject$year_of_birth)
ap <- subject %>%
  filter(infancy_vac == "aP")
round(summary(time_length(ap$age, "years")))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	27	28	28	29	35

```
wp <- subject %>%
  filter(infancy_vac == "wP")
round(summary(time_length(wp$age, "years")))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	33	35	37	40	58

Q8. Determine the age of all individuals at time of boost?

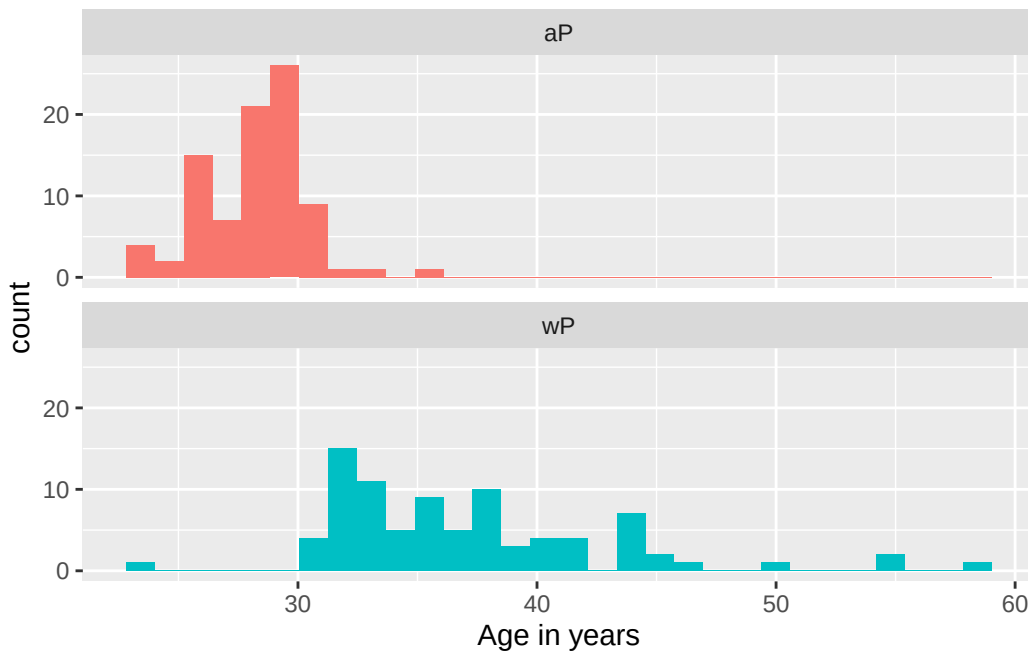
```
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

```
ggplot(subject) +
  aes(time_length(age, "year"),
      fill=as.factor(infancy_vac)) +
  geom_histogram(show.legend=FALSE) +
  facet_wrap(vars(infancy_vac), nrow=2) +
  xlab("Age in years")
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



I think the two groups are significantly different. The two groups clearly lie on different parts of the graph, with wP being more towards the middle than aP.

Joining multiple tables

```
# Complete the API URLs...
specimen <- read_json("https://www.cmi-pb.org/api/specimen", simplifyVector = TRUE)
titer <- read_json("https://www.cmi-pb.org/api/plasma_ab_titer", simplifyVector = TRUE)
```

Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

```
meta <- inner_join(specimen, subject)
```

Joining with ``by = join_by(subject_id)``

```
dim(meta)
```

```
[1] 1503  14
```

```
head(meta)
```

```
specimen_id subject_id actual_day_relative_to_boost
1           1           1                      -3
2           2           1                       1
3           3           1                       3
4           4           1                       7
5           5           1                      11
6           6           1                      32
planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
1                             0         Blood     1         wP         Female
2                             1         Blood     2         wP         Female
3                             3         Blood     3         wP         Female
4                             7         Blood     4         wP         Female
5                            14         Blood     5         wP         Female
6                            30         Blood     6         wP         Female
ethnicity race year_of_birth date_of_boost dataset
1 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
2 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
3 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
4 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
5 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
6 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
age
1 14761 days
2 14761 days
3 14761 days
4 14761 days
5 14761 days
6 14761 days
```

Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
abdata <- inner_join(titer, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```
dim(abdata)
```

```
[1] 52576    21
```

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```
table(abdata$isotype)
```

```
 IgE   IgG  IgG1  IgG2  IgG3  IgG4
6698  5389 10117 10124 10124 10124
```

Q12. What are the different \$dataset values in abdata and what do you notice about the number of rows for the most “recent” dataset?

```
table(abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
          31520          8085          7301          5670
```

The most recent dataset has the least amount.

Examine IgG Ab titer levels

```
igg <- abdata %>% filter(isotype == "IgG")
head(igg)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgG	TRUE	PT	68.56614	3.736992
2	1	IgG	TRUE	PRN	332.12718	2.602350
3	1	IgG	TRUE	FHA	1887.12263	34.050956
4	19	IgG	TRUE	PT	20.11607	1.096366
5	19	IgG	TRUE	PRN	976.67419	7.652635
6	19	IgG	TRUE	FHA	60.76626	1.096457

	unit	lower_limit_of_detection	subject_id	actual_day_relative_to_boost
1	IU/ML	0.530000	1	-3
2	IU/ML	6.205949	1	-3

3	IU/ML	4.679535	1	-3
4	IU/ML	0.530000	3	-3
5	IU/ML	6.205949	3	-3
6	IU/ML	4.679535	3	-3

	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex
1	0	Blood	1	wP	Female
2	0	Blood	1	wP	Female
3	0	Blood	1	wP	Female
4	0	Blood	1	wP	Female
5	0	Blood	1	wP	Female
6	0	Blood	1	wP	Female

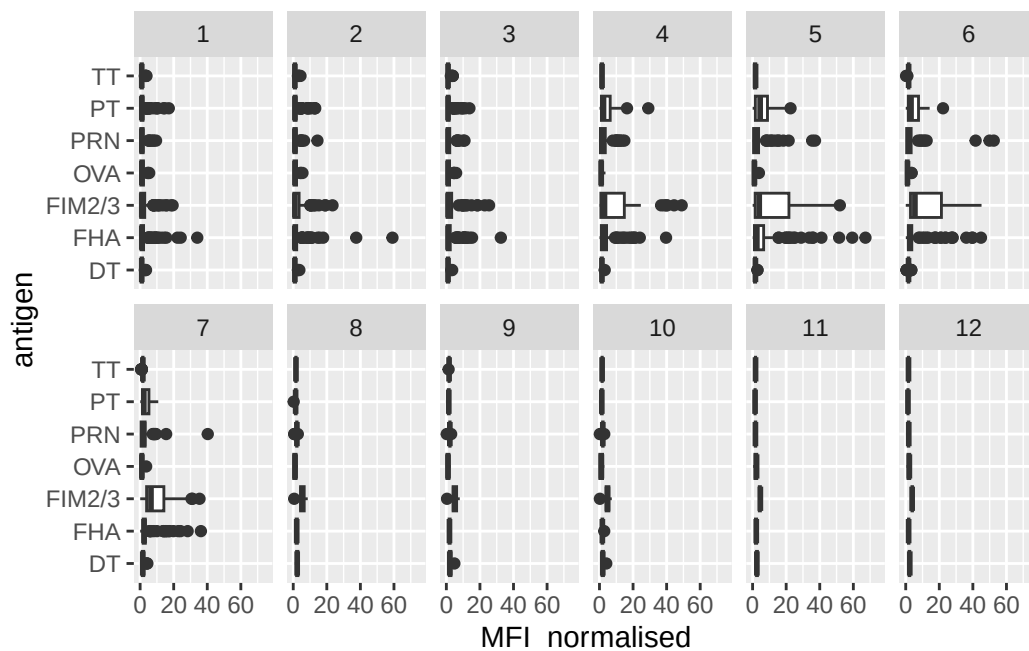
	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Unknown	White	1983-01-01	2016-10-10	2020_dataset
5	Unknown	White	1983-01-01	2016-10-10	2020_dataset
6	Unknown	White	1983-01-01	2016-10-10	2020_dataset

	age
1	14761 days
2	14761 days
3	14761 days
4	15857 days
5	15857 days
6	15857 days

Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
  xlim(0,75) +
  facet_wrap(vars(visit), nrow=2)
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).

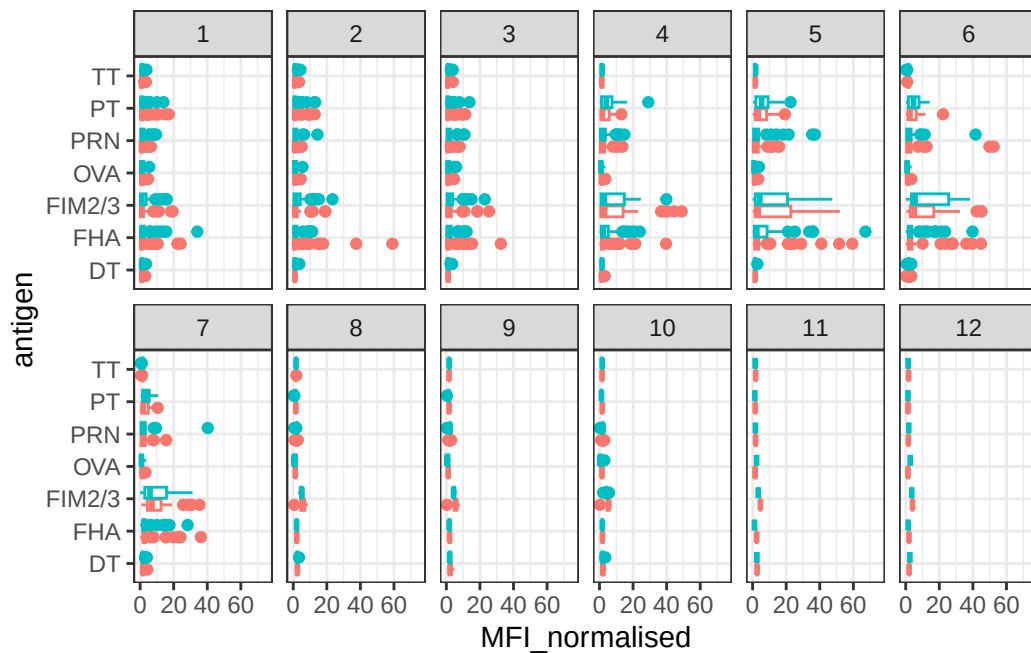


Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

FIM2/3 shows differences in the level of IgG antibody titers. This may be due to how the vaccine may target different antibodies, and in this case, Fim2/3 is targeted for immune memory.

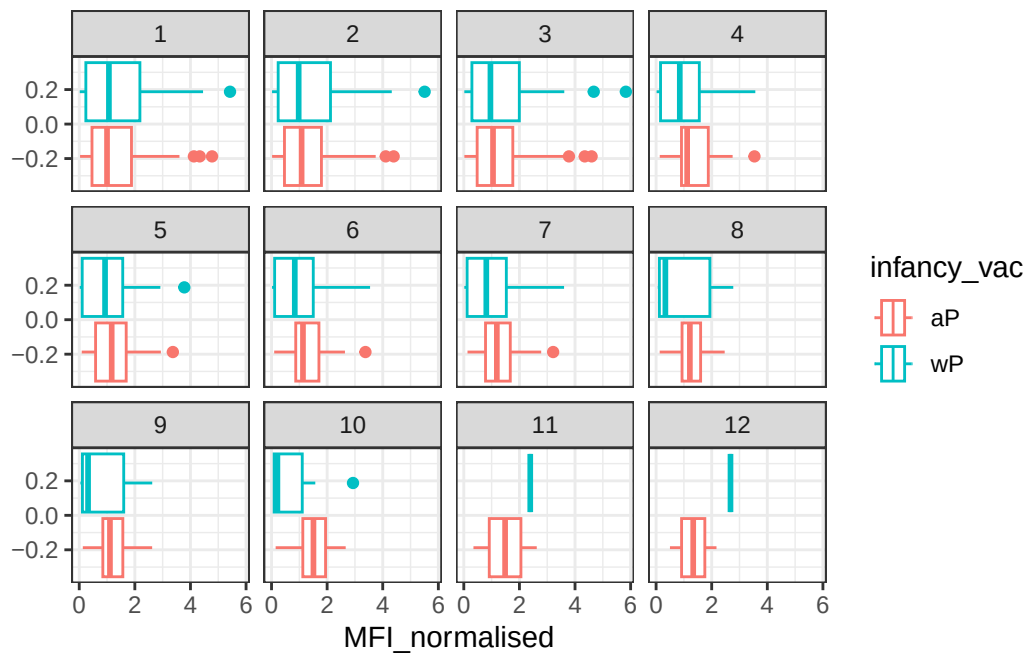
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).

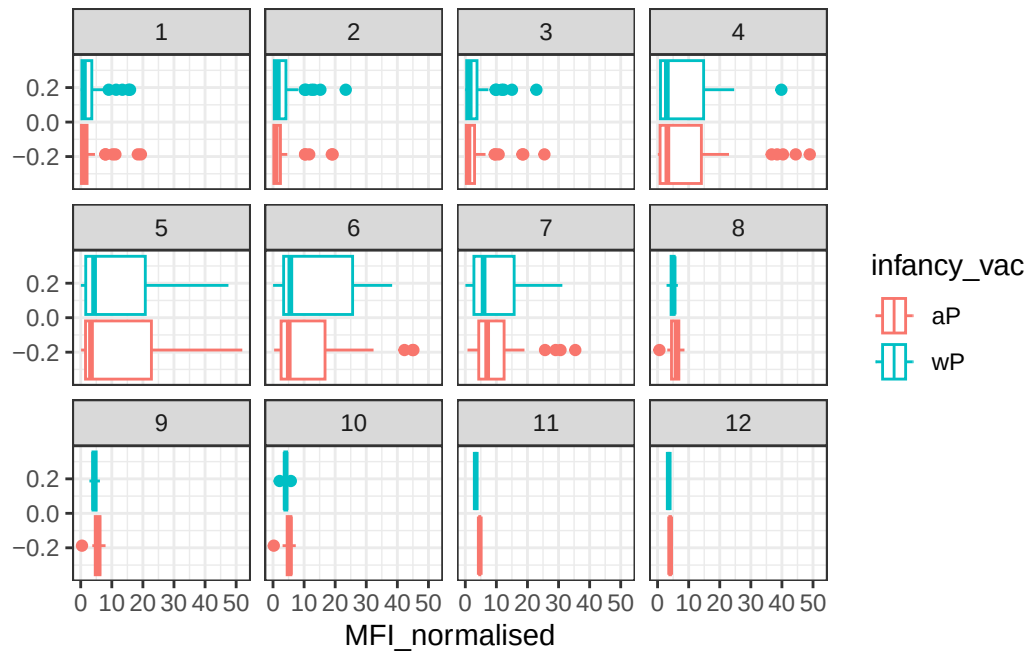


Q15. Filter to pull out only two specific antigens for analysis and create a boxplot for each. You can chose any you like. Below I picked a “control” antigen (“OVA”, that is not in our vaccines) and a clear antigen of interest (“PT”, Pertussis Toxin, one of the key virulence factors produced by the bacterium *B. pertussis*).

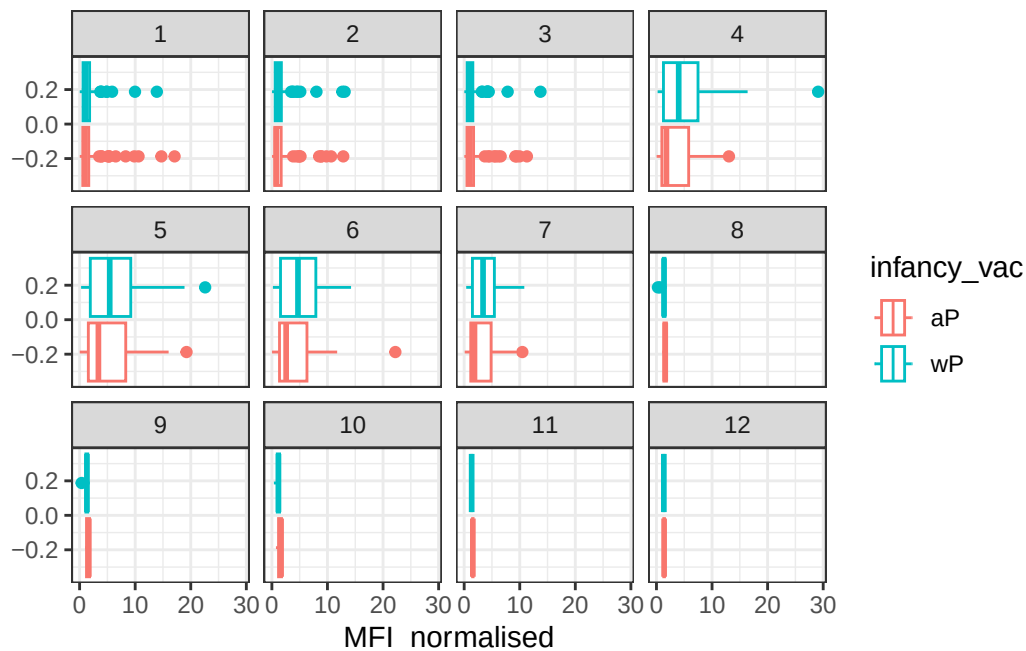
```
filter(igg, antigen=="OVA") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = T) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



```
filter(igg, antigen=="FIM2/3") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = T) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



```
filter(igg, antigen=="PT") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = T) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



Q16. What do you notice about these two antigens time courses and the PT data in particular?

The OVA antigen does not change much overtime. However, the PT and FIM2/3 antibodies show a clear rise and decline over time, for both wP and aP subjects.

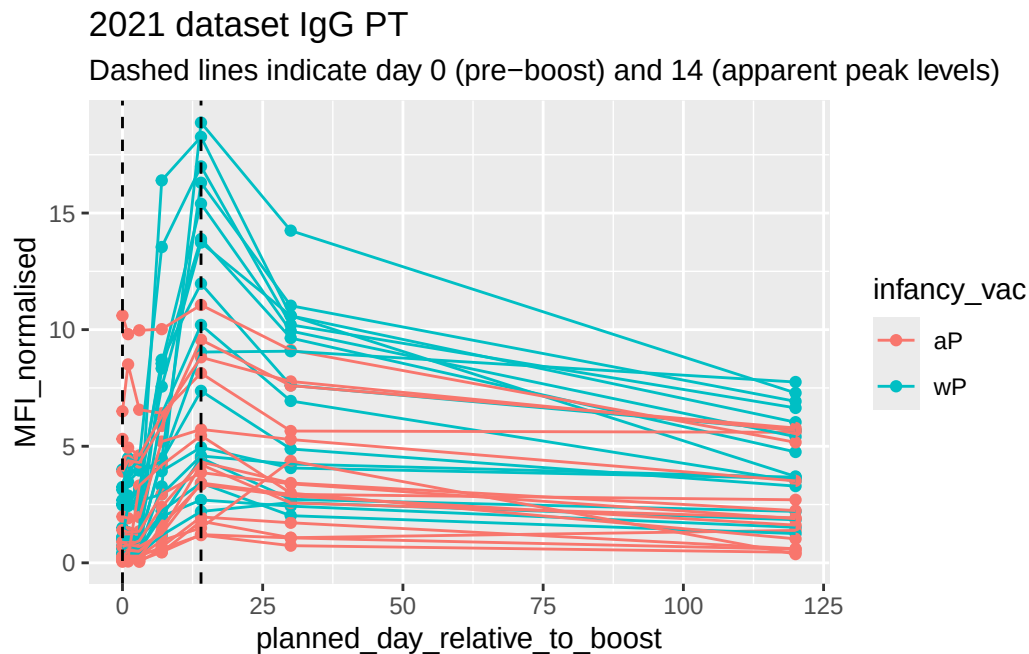
Q17. Do you see any clear difference in aP vs. wP responses?

There does not seem to be a huge difference between the two types.

```
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
         y=MFI_normalised,
         col=infancy_vac,
         group=subject_id) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept=0, linetype="dashed") +
  geom_vline(xintercept=14, linetype="dashed") +
```

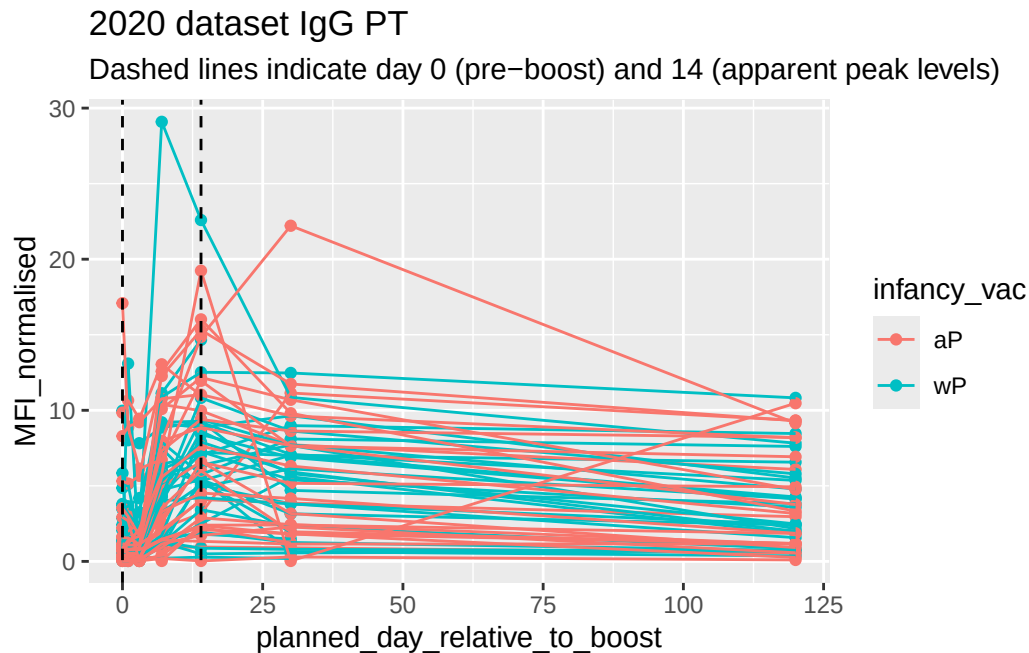
```
labs(title="2021 dataset IgG PT",
      subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```



Q18. Does this trend look similar for the 2020 dataset?

```
abdata.20 <- abdata %>% filter(dataset == "2020_dataset")

abdata.20 %>%
  filter(isotype == "IgG", antigen == "PT", planned_day_relative_to_boost < 200) %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
         y=MFI_normalised,
         col=infancy_vac,
         group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
    labs(title="2020 dataset IgG PT",
          subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```



Compared to the 2020 data, the trend looks similar, jumping up in MFI at the dashed, and then stabilizing after.

Obtaining CMI-PB RNASeq data

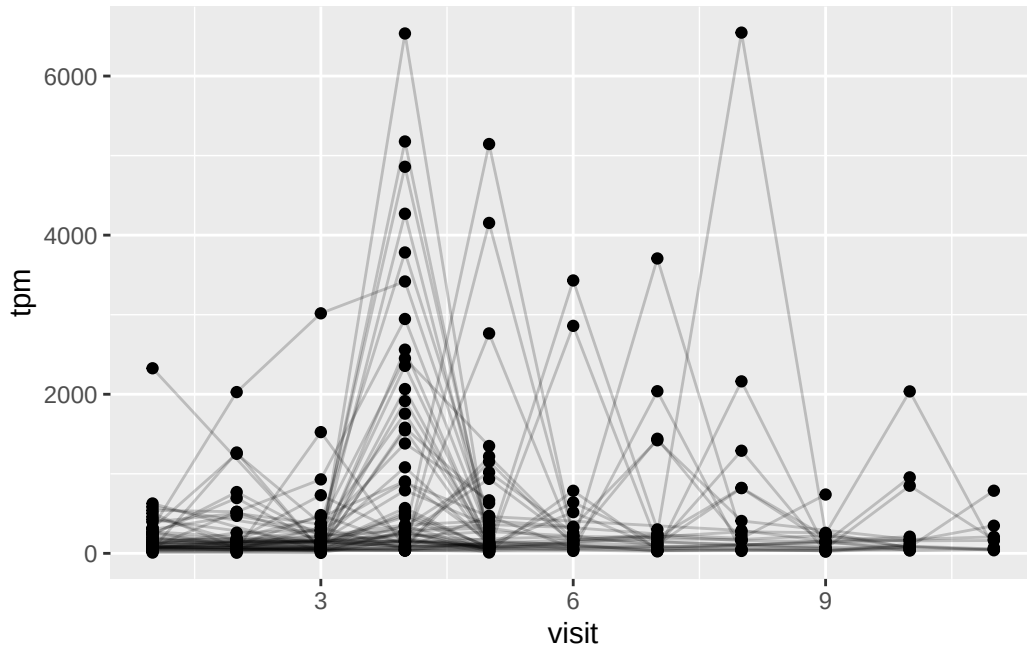
```
url <- "https://www.cmi-pb.org/api/v2/rnaseq?versioned_ensembl_gene_id=eq.ENSOG00000211896.7"
rna <- read_json(url, simplifyVector = TRUE)
```

```
#meta <- inner_join(specimen, subject)
ssrna <- inner_join(rna, meta)
```

Joining with ``by = join_by(specimen_id)``

Q19. Make a plot of the time course of gene expression for IGHG1 gene (i.e. a plot of visit vs. tpm).

```
ggplot(ssrna) +
  aes(visit, tpm, group=subject_id) +
  geom_point() +
  geom_line(alpha=0.2)
```

Q20.: What do you notice about the expression of this gene (i.e. when is it at it's maximum level)?

The expression of the gene is at its highest on visit 4.

Q21. Does this pattern in time match the trend of antibody titer data? If not, why not?

The pattern in time does match the trend of the antibody titer, as the titers that saw significant change did so in visit 4.